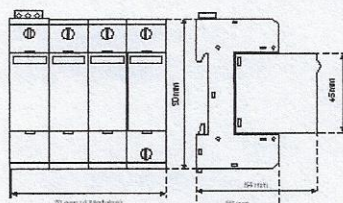
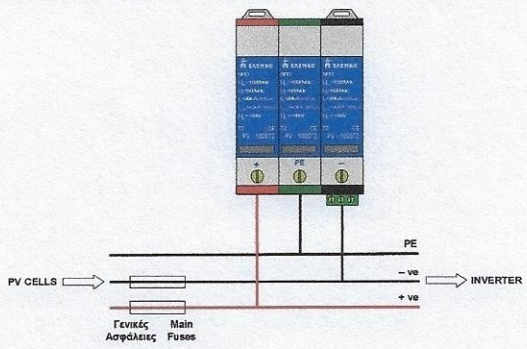


| <b>SURGE PROTECTIVE DEVICES (SPDs)</b><br><b>THREE PHASE 40T2, 230V / 50Hz PROTECTION TYPE T2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Part number ELEMKO : 68 44 448</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                       |
| <b>Introduction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |
| <p>Three phase SPDs 40T2 containing metal oxide varistors (with pluggable protection module) having a response time of less than 25ns and therefore having a low voltage protection level. The SPDs 40T2 limiting type have successfully passed the requirements of European standard EN 61653 – 11 and international IEC 61643 – 1 for class II and are suitable for type T2 providing reliable protection to electrical and electronic systems of insulation level, category IV up to category II according to IEC 60364-4-43. The three phase 40T2 SPDs (3+1) – ELEMKO 68 44 447 &amp; 68 44 448) are suitable for installation in TN and TT systems. They provide secondary protection or primary protection in areas with low lightning density or in buildings where external Lightning Protection System (LPS) is absent. They are mainly installed at the main switch board at the entrance of the distribution network 230/400V, 50Hz and they are recommended for the protection of pump stations, country houses, industries, mobile and stationary telecom facilities, radar, TV – radio stations etc.</p> |                                                                                       |
| <b>Technical data</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                       |
| Number of Poles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4                                                                                     |
| Connection between terminals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $L_1 - N, L_2 - N, L_3 - N \text{ \& } N - PE$                                        |
| Nominal voltage of the system $U_N$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 230/400V, 50Hz                                                                        |
| Maximum operating voltage, $U_c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 440V, 50Hz / 580V DC                                                                  |
| Maximum continuous operating voltage, $U_{MOOV}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 440V, 50Hz / 580V DC                                                                  |
| $I_{max}$ , "class II" test, (8/20 $\mu$ s), 1P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 40kA                                                                                  |
| $I_n$ "class II" test, (8/20 $\mu$ s), 1P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 20kA                                                                                  |
| $U_p$ , (at $I_n$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <2,2kV                                                                                |
| TOV Withstand (3+1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1453V, 50Hz                                                                           |
| Main circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 x MOV (L – N) + 1 x GDT (N – PE)                                                    |
| Response time, $t_A$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <25ns (MOV) / <50ns (GDT)                                                             |
| Short circuit withstand, $I_k$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 25kA / 50Hz                                                                           |
| Back up fuse at serial wiring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <125A gL/gG                                                                           |
| Thermal disconnection device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | YES                                                                                   |
| Dimensions (mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 90 x 64 x 70 mm                                                                       |
| Monitoring indication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | YES                                                                                   |
| Conductor terminals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 35 mm <sup>2</sup>                                                                    |
| Insulation resistance (500Vdc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | >1M $\Omega$                                                                          |
| Housing material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Polycarbonate halogen free                                                            |
| Protection level of housing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IP20                                                                                  |
| Operating temperature, $\vartheta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -40°C ÷ +80°C                                                                         |
| Rail mounting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DIN-3 (TS-35/EN50022)                                                                 |
| Standards                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | EN 61643-11, IEC 61643-1                                                              |
| Part number of pluggable protection module code (L – N)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 68 44 420 (40T2)                                                                      |
| Part number of pluggable protection module code (N – PE)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 68 44 428 (40GT2)                                                                     |
| SPD with RC contacts (part number)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 68 44 447                                                                             |
| RC contacts conductor terminals (mm <sup>2</sup> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1,5 mm <sup>2</sup>                                                                   |
| Drawing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Photo                                                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |



| <b>TWO POLE SURGE PROTECTIVE DEVICES (SPDs)</b><br><b>PV -1000 T1™, FOR PHOTOVOLTAIC SYSTEMS PROTECTION TYPE T1+T2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Part number ELEMKO : 68 10 312</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                              |
| <b>Introduction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                              |
| <p>Two pole SPDs PV – 1000 T1 especially designed for surge protection of (DC / AC) inverters, which are widely used in photovoltaic applications. The SPDs PV – 1000 T1 are made out of heavy duty varistor elements and have successfully passed the requirements of European standard EN 61643 – 11 and international IEC 61643 – 1 for class I &amp; II and are suitable for type T1+T2 providing reliable protection to electrical systems of insulation level, category IV up to category III according to IEC 60364-4-44. They are suitable for direct installation exactly at the input terminals of the inverter just after the entrance of the DC current arriving from the photovoltaic panels.</p> |                              |
| <b>Technical data</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                              |
| Number of Poles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2                            |
| Connection between terminals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $L_{+ve} - PE, L_{-ve} - PE$ |
| Nominal voltage of the system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 850V DC                      |
| Maximum operating voltage, $U_c (L - GND)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1000V DC                     |
| Maximum operating voltage, $U_c (L_{+ve} - L_{-ve})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1000V DC                     |
| Maximum operating voltage of module 1P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 500V DC                      |
| $I_{imp}$ , "class I" test, (10/350 $\mu$ s), 1P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 12,5kA                       |
| $I_{max}$ , "class II" test, (8/20 $\mu$ s), 1P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 50kA                         |
| $I_n$ "class II" test, (8/20 $\mu$ s), 1P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 20kA                         |
| $U_p$ , (at $I_n$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <4kV                         |
| Main circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MOV                          |
| Response time, $t_a$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <25ns                        |
| Short circuit withstand, $I_k$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3kA / 50Hz                   |
| Thermal disconnection device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | YES                          |
| Dimensions (mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 90 x 64 x 52,5 mm            |
| Monitoring indication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | YES                          |
| Conductor terminals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 35 mm <sup>2</sup>           |
| Housing material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Polycarbonate halogen free   |
| Protection level of housing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | IP20                         |
| Operating temperature, $\vartheta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -40°C + +80°C                |
| Rail mounting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | DIN-3 (TS-35/EN50022)        |
| Standards                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EN 61643-11, IEC 61643-1     |
| SPD with RC contacts (part number)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>68 10 314</b>             |
| RC contacts conductor terminals (mm <sup>2</sup> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1,5 mm <sup>2</sup>          |
| <b>Drawing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                              |

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